L11 - Divide & Conquer

Tuesday, April 21, 2020 8:09 AM

General Method

```
Algorithm DAC(P)
{
      if Small(P)
         return DS(P)
      else {
          divide P into smaller instances P1, P2,…, Pk, K >= 1;   //step 1  (Divide phase)
          Apply DAC to each of these subproblems;                 // step2
          return Combine(DAC( P1), DAC( P2),…, DAC( Pk));     // step 3 (Combine phase)

      //This above step ("Combine") may not be present in some of the problems that
      fall under this strategy.  Note, return will always executed.
      }
}
```

13 4 5 1 9 15 2 8 11 10

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13 4 | 5 1 9 | 15 2 | 8 11 10

13 | 4 | 5 | 1 9 | 15 | 2 | 8 | 11 10

$$T(p) = \begin{cases} s(p); & \textit{if } p \textit{ is small} \\ T(p_1) + T(p_2) + \cdots + T(p_k) + f(p); & \textit{otherwise} \end{cases}$$